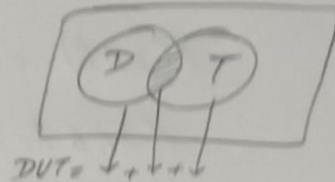


ISM1) $\Rightarrow D = \text{all software developers}$ $T = \text{all software testing engineers}$ Given:

$$P(D \cap T) = 0.25$$

$$P(D - T) = 0.05 = P(D) - P(D \cap T)$$

$$P(T - D) = 0.1 = P(T) - P(D \cap T)$$



$$\text{To find: } P(T \cap D) = P(D \cap T) - P(D - T) - P(T - D)$$

$$= \boxed{0.1}$$

b)

 $H = \text{event that head appears}$ $T = \text{tail}$

$$P(H) = 0.6$$

$$P(T) = 0.4$$

 $X = \text{no of heads appearing}$

$$n = 20$$

$$x = 5$$

$$\text{let } p = P(H) \Rightarrow 1 - p = P(T)$$

 $\Rightarrow \text{Binomial}$

$$P(X=x) = {}^nC_x p^x (1-p)^{n-x}$$

$$P(X=5) = {}^{20}C_5 (0.6)^5 (0.4)^{15} = \boxed{0.00129}$$

Poisson

$$P(X=x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

$$\lambda = np = 12$$

$$P(X=5) = \frac{12^5}{5!} e^{-12} = \boxed{0.01274}$$

Binomial gives more accurate result because of the situation
 (n is not too large, p is not too small).

2) a)

A = patient is selected with high BP

B = " " " diabetic

C = " " " with cancer

$$P(A) = 0.35 \quad P(B) = 0.4 \quad P(C) = 0.45$$

$$P(A \cap B) = 0.22 \quad P(B \cap C) = 0.25 \quad P(A \cap C) = 0.22$$

$$P(A \cap B \cap C) = 0.15$$

$$\begin{aligned} \text{i) } P(A \cup B \cup C) &= 0.35 + 0.4 + 0.45 - 0.22 - 0.25 - 0.22 + 0.15 \\ &= 0.66 \end{aligned}$$

$$\text{ii) } P(A|B \cup C) = \frac{P(A \cap (B \cup C))}{P(B \cup C)} = \frac{P(A \cap B) \cup P(A \cap C)}{P(B \cup C)}$$

let $E = B \cup C$

$$= \frac{P(A \cap E)}{P(E)}$$

$$= \frac{-P(A \cap E) + P(A) + P(E)}{P(E)}$$

$$= \frac{P(B \cup C) + P(A) - P(A \cup B \cup C)}{P(B \cup C)}$$

$$\begin{aligned} P(B \cup C) &= P(B) + P(C) - P(B \cap C) \\ &= 0.4 + 0.45 - 0.25 \\ &= 0.6 \end{aligned} \quad \begin{aligned} &= \frac{0.6 + 0.35 - 0.66}{0.6} \\ &= 0.483 \end{aligned}$$

$$\text{iii) } P(B|A \cup C) = \frac{P(A \cup C) + P(B) - P(A \cup B \cup C)}{P(A \cup C)}$$

$$\begin{aligned} P(A \cup C) &= 0.35 + 0.45 - 0.22 \\ &= 0.58 \end{aligned} \quad \begin{aligned} &= \frac{0.58 + 0.4 - 0.66}{0.58} \\ &= 0.5517 \end{aligned}$$

27b)

A = items produced by operator A

B = " " " " B

C = " " " " C

D = " " " " D

T = defective item is produced

$$P(A) = 0.3$$

$$P(T|A) = 0.01$$

$$P(B) = 0.3$$

$$P(T|B) = 0.05$$

$$P(C) = 0.2$$

$$P(T|C) = 0.06$$

$$P(D) = 0.2$$

$$P(T|D) = 0.07$$

$$P(T) = P(A)P(T|A) + P(B)P(T|B) + P(C)P(T|C) + P(D)P(T|D)$$

$$= 0.044$$

$$i) \quad P(A|T) = \frac{P(T|A)P(A)}{P(T)} = \frac{0.01 \times 0.3}{0.044} = \boxed{0.068}$$

ii) ~~Let~~ Let B_2 be new time

$$P(B_2) = 0.15$$

$$P(T_2|B_2) = 0.05$$

$$\text{new } P(T_2) = 0.0365$$

$$P(B_2|T_2) = \frac{P(T_2|B_2)P(B_2)}{P(T_2)} = \boxed{0.205}$$

3) a) Given:

$$n = 1200$$

$$p = 0.02$$

New:

$$\mu = np = 24$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{1200 \times 0.02 \times 0.98} = 4.8497$$

$$z_{30} = \frac{x - \mu}{\sigma} = \frac{30 - 24}{4.8497} = \boxed{1.2372}$$

$$P(X \geq 30) = 1 - P(z_{30}) = 1 - 0.892 = \boxed{0.108}$$

3) b)

PDF: $f(x) = k(1-x^2) \quad 0 \leq x \leq 2$

i) $\sum_{x=0}^2 f(x) = 1$

$$\Rightarrow k(1-0^2) + k(1-1^2) + k(1-2^2) = 1$$

$$\Rightarrow k + 0 + (-3)k = 1$$

$$\Rightarrow -2k = 1$$

$$k = -\frac{1}{2}$$

$$\int_0^2 f(x) dx = 1$$

$$k \int_0^2 (1-x^2) dx = 1$$

$$k \left[x - \frac{x^3}{3} \right]_0^2 = 1$$

$$\Rightarrow k \left(2 - \frac{8}{3} - 0 \right) = 1$$

$$\Rightarrow k \left(-\frac{2}{3} \right) = 1$$

$$\Rightarrow \boxed{k = -\frac{3}{2}}$$

ii) $P(0.5 \leq X \leq 2) = -\frac{3}{2} \left[x - \frac{x^3}{3} \right]_{0.5}^2$



$$= -\frac{3}{2} \left(\left(2 - \frac{8}{3} \right) - \left(0.5 - \frac{(0.5)^3}{3} \right) \right)$$

$$= \boxed{\frac{27}{16}}$$

iii) $P(X=2) = 0$

$$\left(-\frac{3}{2} \left[x - \frac{x^3}{3} \right]_2^2 = 0 \right)$$

4) a)

$$f(x, y) = \begin{cases} k(2x+y) & 2 < x < 5, 0 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

$$i) \int_2^5 \int_0^4 k(2x+y) dy dx = 1$$

$$\Rightarrow k \int_2^5 \left[2xy + \frac{y^2}{2} \right]_0^4 dx = 1$$

(partial integration)

$$\Rightarrow k \int_2^5 \left(8x + \frac{16}{2} \right) dx = 1$$

$$\Rightarrow k \int_2^5 (8x + 8) dx = 1$$

$$\Rightarrow 8k \left[\frac{x^2}{2} + x \right]_2^5 = 1$$

$$\Rightarrow 8k \left(\left(\frac{25}{2} + 5 \right) - \left(\frac{4}{2} + 2 \right) \right) = 1$$

$$\Rightarrow 8k \left(\frac{27}{2} \right) = 1$$

$$\Rightarrow 108k = 1 \Rightarrow \cancel{k = 1/108}$$

$$\boxed{k = \frac{1}{108}}$$

$$ii) P(x > 2, y > 3) = \frac{1}{108} \int_2^5 \int_3^4 (2x+y) dy dx \quad \begin{pmatrix} x < 5 \\ y < 4 \end{pmatrix}$$

$$= \frac{1}{108} \int_2^5 \left[2xy + \frac{y^2}{2} \right]_3^4 dx$$

$$= \frac{1}{108} \int_2^5 \left(2x + \frac{7}{2} \right) dx$$

$$= \frac{1}{108} \left[x^2 + \frac{7x}{2} \right]_2^5$$

$$= \frac{1}{108} \left(21 + \frac{21}{2} \right) = \boxed{\frac{7}{24}}$$

iii)

$$P(X > 4) = \frac{1}{108} \int_0^4 \int_4^5 (2x+y) dx dy$$

$$\begin{pmatrix} x < 5 \\ 0 < y < 4 \end{pmatrix}$$

$$= \frac{1}{108} \int_0^4 [x^2 + xy]_4^5 dy$$

$$= \frac{1}{108} \int_0^4 (9+y) dy$$

$$= \frac{1}{108} \left[9y + \frac{y^2}{2} \right]_0^4$$

$$= \frac{1}{108} (36+8) = \boxed{\frac{11}{27}}$$

iv)

$$(X+Y) < 3$$

$$X > 2 \Rightarrow Y = 0 \Rightarrow \min$$

$$Y = 1 \Rightarrow \max$$

$$Y > 0 \Rightarrow X = 3 \Rightarrow \max$$

$$X = 2 \Rightarrow \min$$

$$(2,0), (2,1), (3,0)$$

$$y = 3-x$$

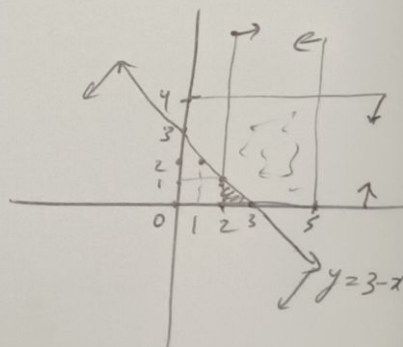


$$\frac{1}{108} \int_2^3 \int_0^{3-x} (2x+y) dy dx$$

$$\Rightarrow \frac{1}{108} \int_2^3 \left[2xy + \frac{y^2}{2} \right]_0^{3-x} dx$$

$$= \frac{1}{108} \int_2^3 \left(2x(3-x) + \frac{(3-x)^2}{2} \right) dx$$

$$= \frac{1}{108} \int_2^3 \left(6x - 2x^2 + \frac{9+x^2-6x}{2} \right) dx$$



$$= \frac{1}{108} \left[3x^2 - \frac{2x^3}{3} + \frac{9x}{2} + \frac{x^3}{6} - \frac{3x^2}{2} \right]_2^3$$

$$= \frac{1}{108} \left(3 \times 5 - \frac{2}{3}(19) + \frac{9}{2} + \frac{19}{6} - \frac{3 \times 5}{2} \right)$$

$$= \frac{5}{216} = \boxed{0.0231}$$

4/b)

$P(x,y)$	x		
y		2	4
	1	0.1	0.15
	3	0.2	0.3
	5	0.1	0.15

i) ~~$P(x)$~~
marginal

x	2	4
$P(x)$	0.4	0.6

y	1	3	5
$P(y)$	0.25	0.5	0.25

$$\text{ii) } P(Y=3|X=4) = \frac{P(Y=3, X=4)}{P(X=4)} = \frac{0.3}{0.6} = \boxed{0.5}$$

$$\text{iii) } P(X=2, Y=1) = 0.1$$

$$P(X=2) = 0.4$$

$$P(Y=1) = 0.25$$

$$P(X=2)P(Y=1) = 0.1$$

$$\text{similarly, } P(X=x, Y=y) = P(X=x)P(Y=y)$$

$\therefore$ its independent

$$5) a) N = 1000$$

$$n = 5$$

$$p = \frac{1}{2} \text{ (either boy or girl)}$$

$$q = \frac{1}{2}$$

$X = \text{no of boys}$

(binomial distribution)

$$i) P(X=2) = {}^5C_2 \left(\frac{1}{2}\right)^5 = \frac{5}{16} = \boxed{0.3125} \times 1000 \approx \boxed{312 \text{ families}}$$

$$ii) P(X \geq 1) = 1 - P(X=0) = 1 - {}^5C_0 \left(\frac{1}{2}\right)^5 = \frac{31}{32} = \boxed{0.969} \times 1000 = \boxed{969 \text{ families}}$$

iii) max 3 girls $\Rightarrow$ min 2 boys

$$P(X \geq 2) = 1 - P(X=0) - P(X=1)$$

$$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^5 - {}^5C_1 \left(\frac{1}{2}\right)^5$$

$$= 1 - (0.5)^5 - 5 \times (0.5)^5$$

$$= \frac{13}{16} = \boxed{0.813} \times 1000 = \boxed{813 \text{ families}}$$

iv) $P(X \neq 0 \& X \neq 5)$

$$= 1 - P(X=0) - P(X=5)$$

$$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^5 - {}^5C_5 \left(\frac{1}{2}\right)^5$$

$$= \frac{15}{16} = \boxed{0.938} = \boxed{938 \text{ families}}$$

b) $n = 100$

$$\mu = 12$$

$$\sigma = 3$$

normal distribution

$$\left(Z = \frac{x - \mu}{\sigma} \right)$$

$$i) P(X > 15) = 1 - P\left(Z < \frac{15-12}{3}\right)$$

$$= 1 - P(Z < 1)$$

$$= 1 - 0.8413$$

$$= 0.1587 = \boxed{15.87\%}$$

$$\begin{aligned}
 \text{ii)} \quad P(X < 6) &= P\left(Z < \frac{6-12}{3}\right) \\
 &= P(Z < -2) \\
 &= 0.0227 = \boxed{2.27\%}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii)} \quad P(10 < X < 14) &= P\left(Z < \frac{14-12}{3}\right) - P\left(Z < \frac{10-12}{3}\right) \\
 &= P(Z < 0.66) - P(Z < -0.66) \\
 &= \cancel{0.4950} = \boxed{49.5\%} \\
 &= 0.7475 - 0.2525 \\
 &= 0.495 = \boxed{49.5\%}
 \end{aligned}$$